ANJALI SAINI

+91-9540123502 | anjalisaini2002@gmail.com | Gurugram, Haryana | LinkedIn| GitHub| Website

PROFESSIONAL SUMMARY

M.Sc. Bioinformatics graduate with hands-on experience in computational biology, RNA-seq analysis, molecular docking, and machine learning applications in life sciences. Skilled in Python, R, SQL, Linux, Snakemake, Galaxy, HISAT2, DESeq2, AutoDock, UCSF Chimera, and AlphaFold. Completed a Computational Biology internship involving large-scale pharmaceutical data extraction and processing of 50,000+ drug-label records using Python-based workflows. Experienced in bioinformatics pipeline development, genomic data analysis, and in-silico research for precision medicine and drug discovery.

EDUCATION

M.Sc. Bioinformatics — Maharshi Dayanand University, Rohtak (2024–2026)

B.Sc. Biotechnology — Dronacharya Govt. College, Gurugram (2021–2024)

WORK EXPERIENCE

Computational Biology Intern — Growdea Technologies Pvt. Ltd. | Dec 2025 – Jan 2026

• Processed and structured 50,000+ pharmaceutical records using Python (pandas) and regex-based text mining, improving data readiness by ~40%.

• Managed 3+ in-silico pipelines, enhancing data processing efficiency by 30%.

DISSERTATION RESEARCH

In Silico Evaluation of Phytochemicals Against Antibiotic-Resistant *Klebsiella pneumonia* | M.Sc. Dissertation | 2025–2026

• Performed molecular docking of 102 phytochemicals (filtered from 36,464 compounds) against PBP targets using AutoDock and Raccoon; identified top 25 candidates based on binding affinity.

• Prepared and validated protein structures from PDB and AlphaFold; evaluated drug-likeness using ADMET and Lipinski's Rule of 5, reducing candidates by ~70%.

• Conducted structural and comparative analysis using UCSF Chimera and PyMOL; generated reports with docking visualizations for research interpretation.

Applied protein structure validation, active-site identification, interaction analysis, and ADMET profiling using AlphaFold2, UCSF Chimera, AutoDock 4.2, RACCOON, LigPlot+, and Discovery Studio.

ACADEMIC PROJECTS

Automated RNA-seq Analysis Pipeline using Snakemake and Galaxy | GitHub

- Developed an automated RNA-seq analysis pipeline for E. coli transcriptome data using Snakemake, Galaxy, Linux, and R. Integrated FastQC, Trimmomatic, HISAT2, featureCounts, and DESeq2 to perform quality control, alignment, differential expression analysis, and volcano plot visualization.

WGS and Comparative AMR of Carbapenem-Resistant *Klebsiella pneumoniae* | Galaxy| GitHub

- Performed whole genome sequencing and resistome analysis of isolates from India, Turkey, and the United States using Galaxy Europe. tools including FastQC, Trimmomatic, BWA-MEM2, FreeBayes, and ABRicate. Identified 30,261 genomic variants and region-specific carbapenemase genes such as blaNDM-1, blaKPC-2, and blaKPC-3.

Machine Learning Pipeline — Biomedical Text Analytics | Orange Data Mining| GitHub

- Developed end-to-end machine learning workflows for biomedical text preprocessing, classification, and interpretation; achieved 85%+ accuracy and improved model performance through optimized feature selection.

Machine Learning on Diabetes Dataset | WEKA | **GitHub**

- Implemented classification algorithms (Naive Bayes, Decision Tree, SVM) on a 768-record diabetes dataset; performed EDA and feature engineering, improving prediction accuracy by 12%.

Drug Discovery & Molecular Docking | HEX

- Designed molecular docking workflow using HEX; analyzed protein–ligand interactions and prepared chemical structures using UCSF Chimera and ChemSketch.

Hospital Overview Dashboard | Power BI

- Built a 3-page interactive Power BI dashboard on a healthcare dataset, analyzing 200+ appointments and ₹551K revenue; enabled faster decision-making through KPI visualization and trend analysis.

Python Programming Projects | GitHub Portfolio

- Developed multiple command-line Python applications, including Blackjack, Hangman, Calculator, Secret Auction, and Password Generator. Applied core programming concepts such as functions, loops, conditional statements, dictionaries, and modular code organization

CERTIFICATIONS & ACADEMIC ACTIVITIES

AI & Machine Learning in Pharmaceutical Sciences Certificate course — Maharshi Dayanand University, Rohtak

Academic Poster Presentation on NGS Workflows — Maharshi Dayanand University, Rohtak

Certification on R Programming — Simplilearn

Certification on SQL Bootcamp 2025 — Udemy

Introduction to Cloud Computing — Online Certification

Hands-on Workshop on Bioinformatics & Molecular Docking BioSoc-DTU

Tata Group Data Analytics Job Simulation — Forage

Deloitte Australia Data Analytics Job Simulation — Forage

McKinsey Forward Program — McKinsey & Company

SKILLS

Bioinformatics & Genomics

RNA-seq Analysis, NGS Workflows, Differential Gene Expression Analysis, Molecular Docking, Virtual Screening, Protein–Ligand Interaction Analysis, Functional Annotation, Pharmacogenomics, Biological Data Interpretation

Computational & Programming

Python, R, SQL, Perl, C, Linux, Snakemake, Galaxy, pandas, Scikit-learn, Pandas, Workflow Automation

Bioinformatics Tools & Platforms

FastQC, Trimmomatic, HISAT2, featureCounts, DESeq2, BWA-MEM2, FreeBayes, ABRicate, BLAST, AutoDock, UCSF Chimera, PyMOL, AlphaFold, Discovery Studio, LigPlot+, WEKA, Orange Data Mining, Power BI

Research & Professional Skills

Scientific Literature Review, Biological Data Analysis, Data Visualization, Research Documentation, Presentation Skills, Problem-Solving, Team Collaboration, Analytical Thinking

LANGUAGES

English (Professional) | Hindi (Native)